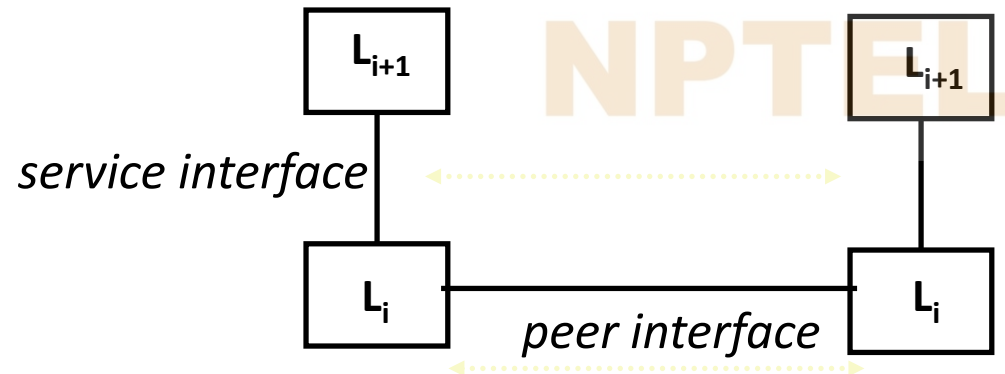
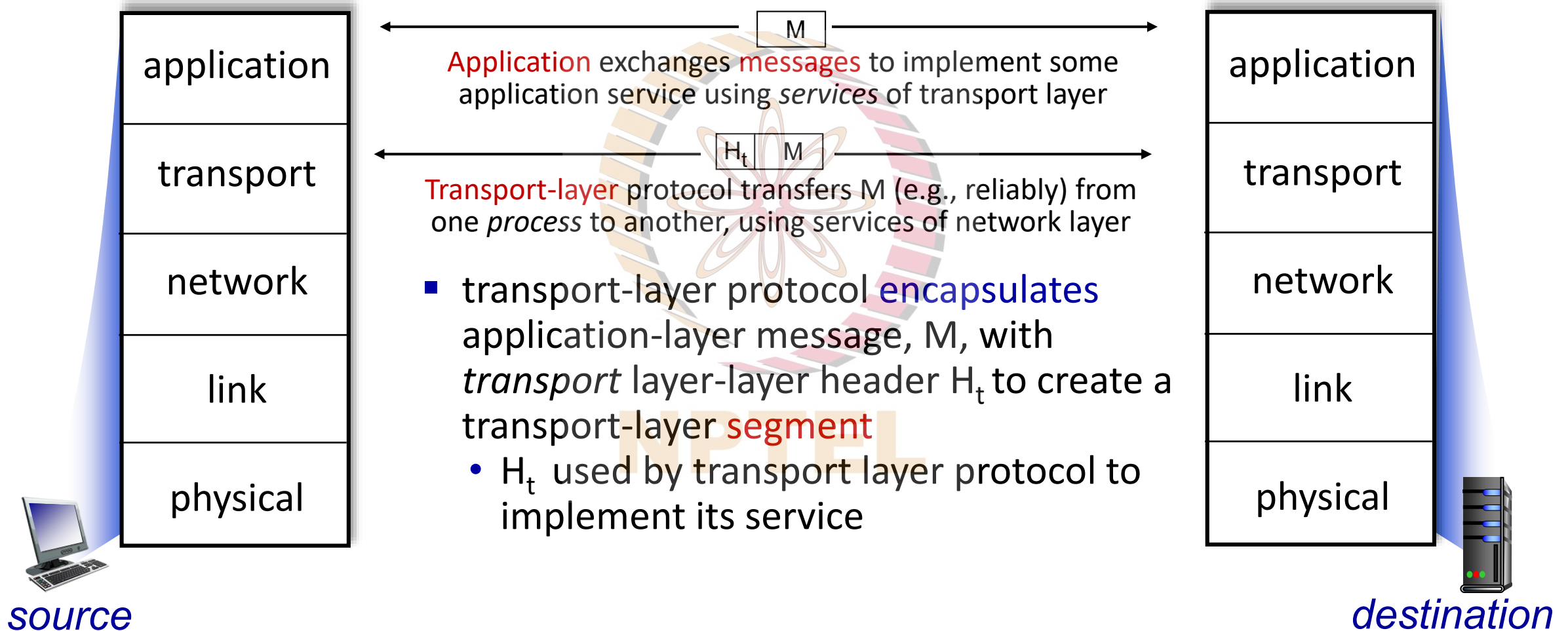


Protocol Implementations

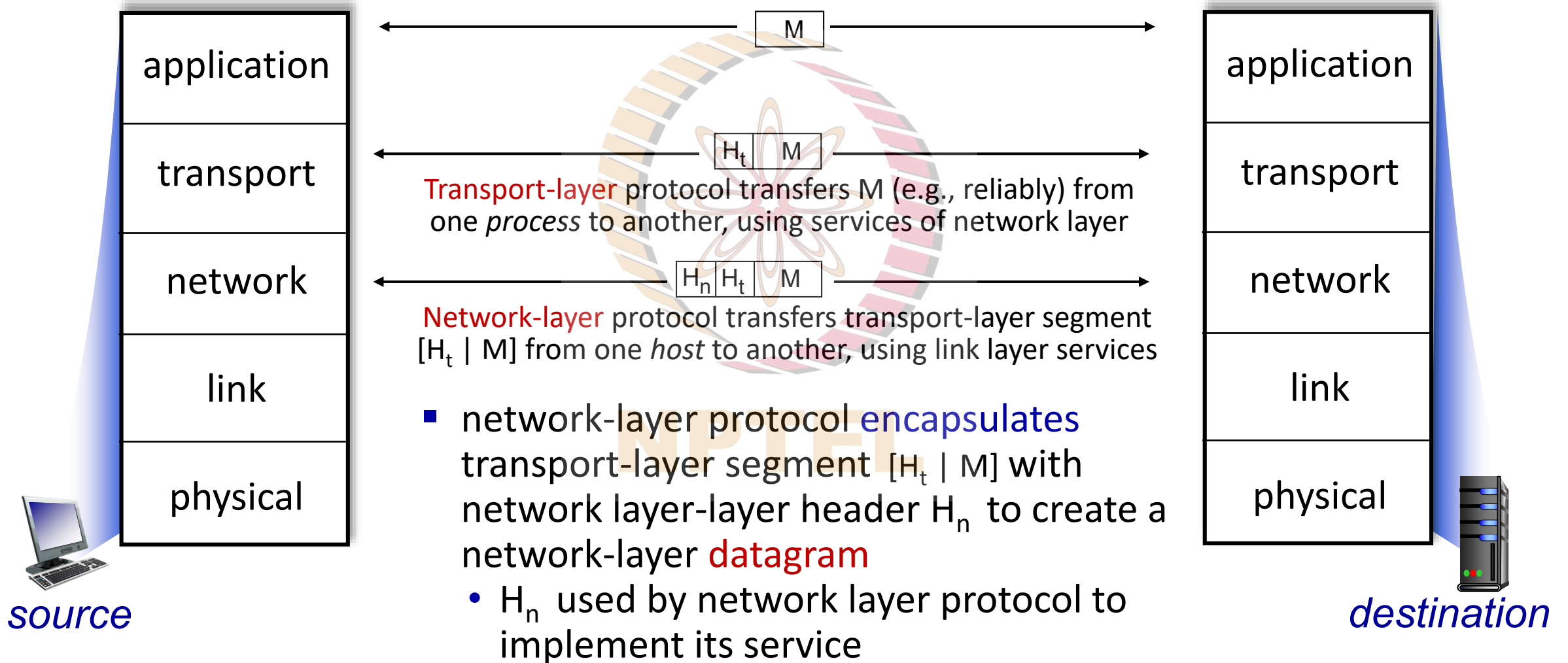
- ❑ Are building blocks of a network architecture
- ❑ Each protocol object has two different interfaces
 - ❑ service interface: defines operations on this protocol
 - ❑ peer-to-peer interface: defines messages exchanged with peer



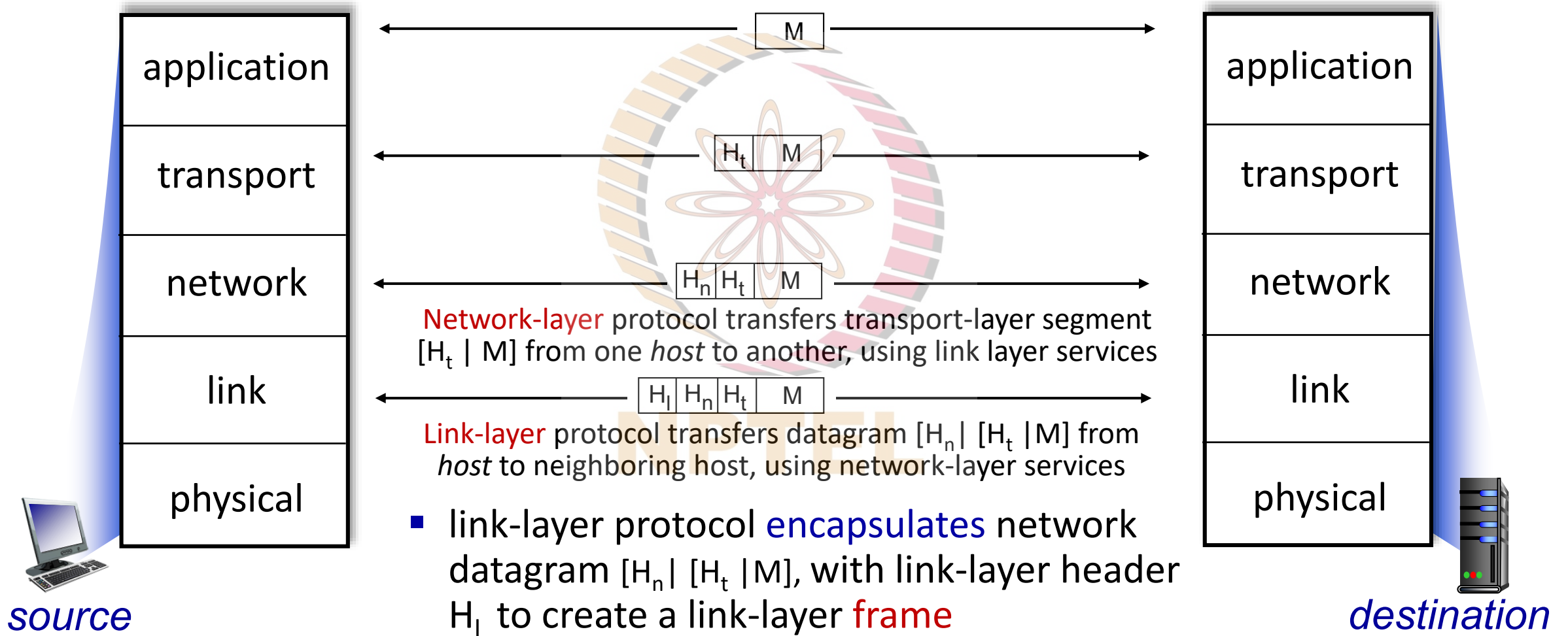
Services, Layering and Encapsulation



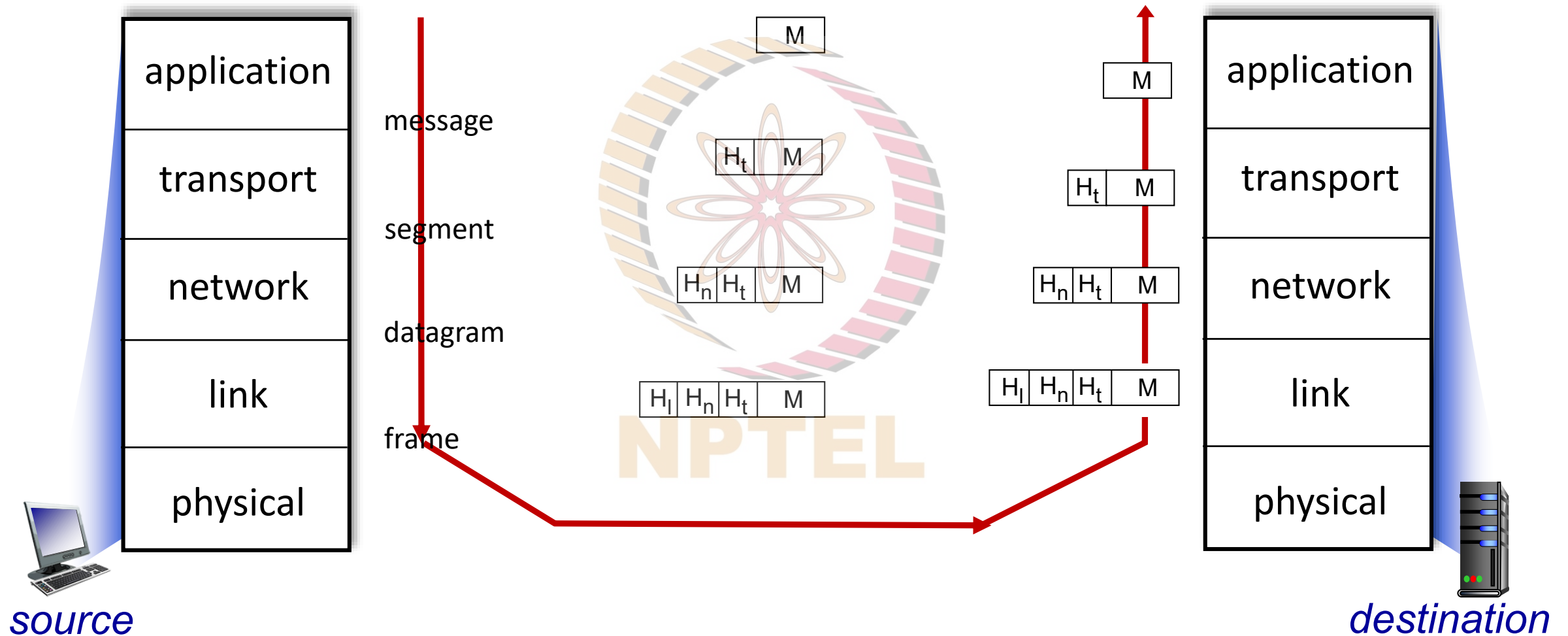
Services, Layering and Encapsulation



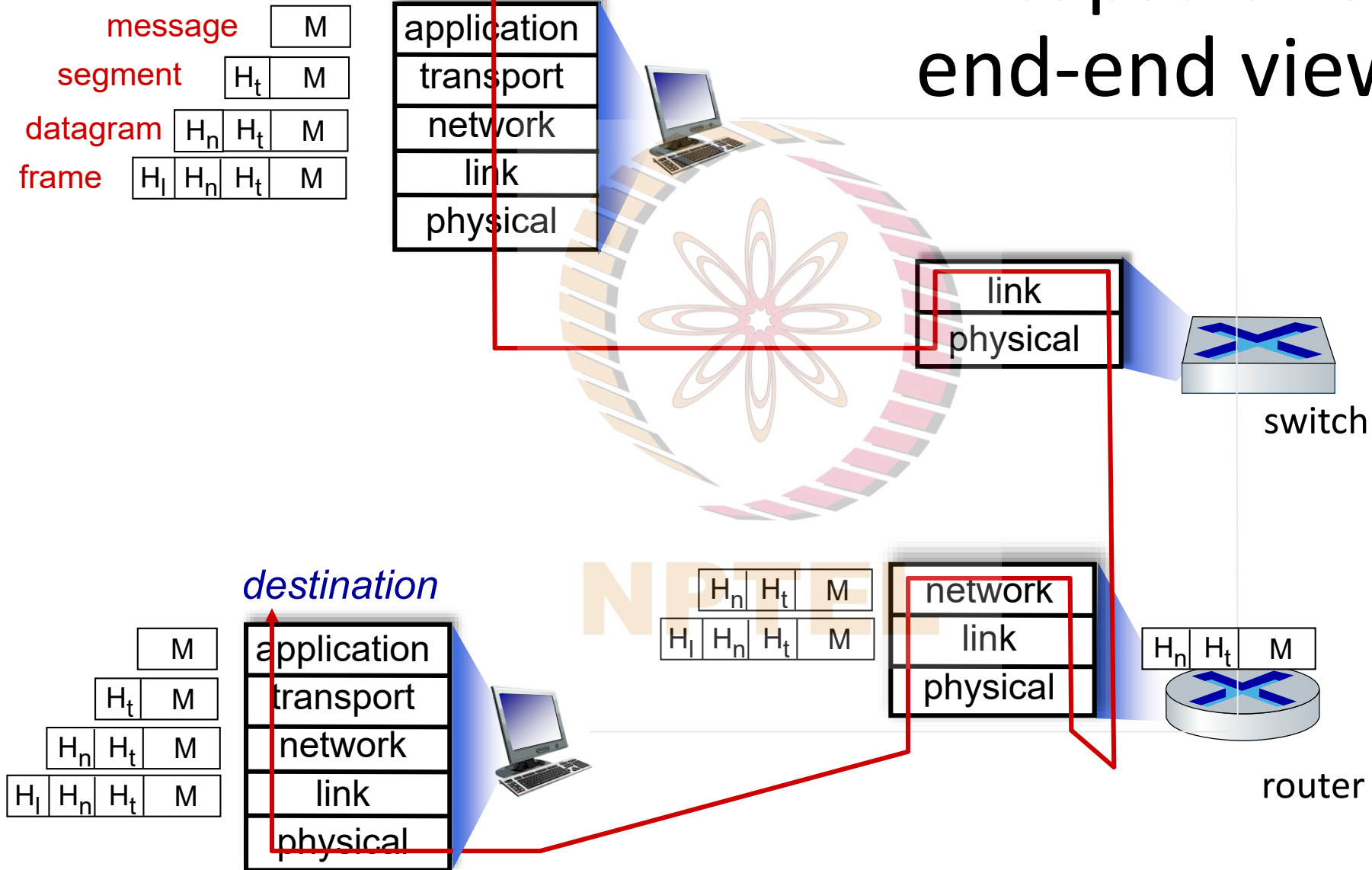
Services, Layering and Encapsulation



Services, Layering and Encapsulation



Encapsulation: an end-end view



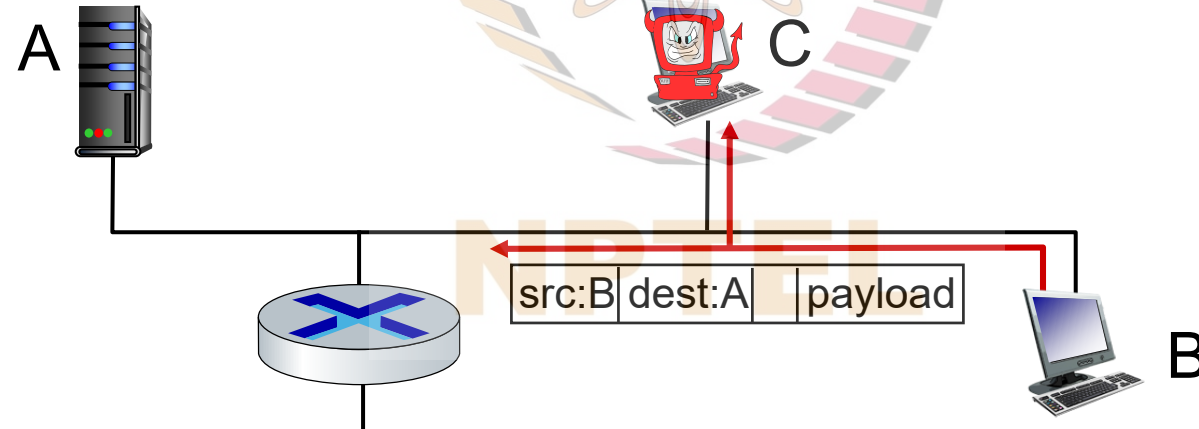
Network security

- ❑ Internet is not originally designed with (much) security in mind
 - ❑ *original vision*: “a group of mutually trusting users attached to a transparent network” 😊
 - ❑ Internet protocol designers playing “catch-up”
 - ❑ security considerations in all layers!
- ❑ We now need to think about:
 - ❑ how bad guys can attack computer networks
 - ❑ how we can defend networks against attacks
 - ❑ how to design architectures that are immune to attacks

Bad guys: Packet Interception

- ❑ *packet “sniffing”*:

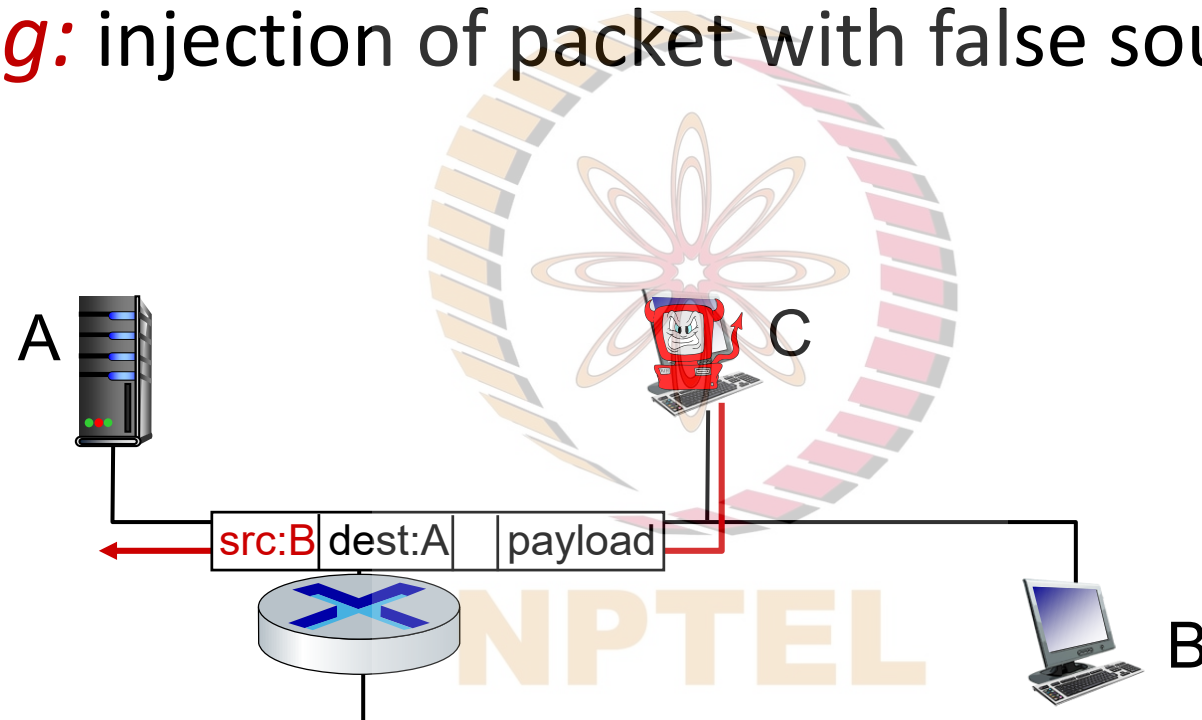
- ❑ broadcast media (shared Ethernet, wireless)
- ❑ promiscuous network interface reads/records all packets (e.g., including passwords!) passing by



Wireshark

Bad guys: Fake Identity

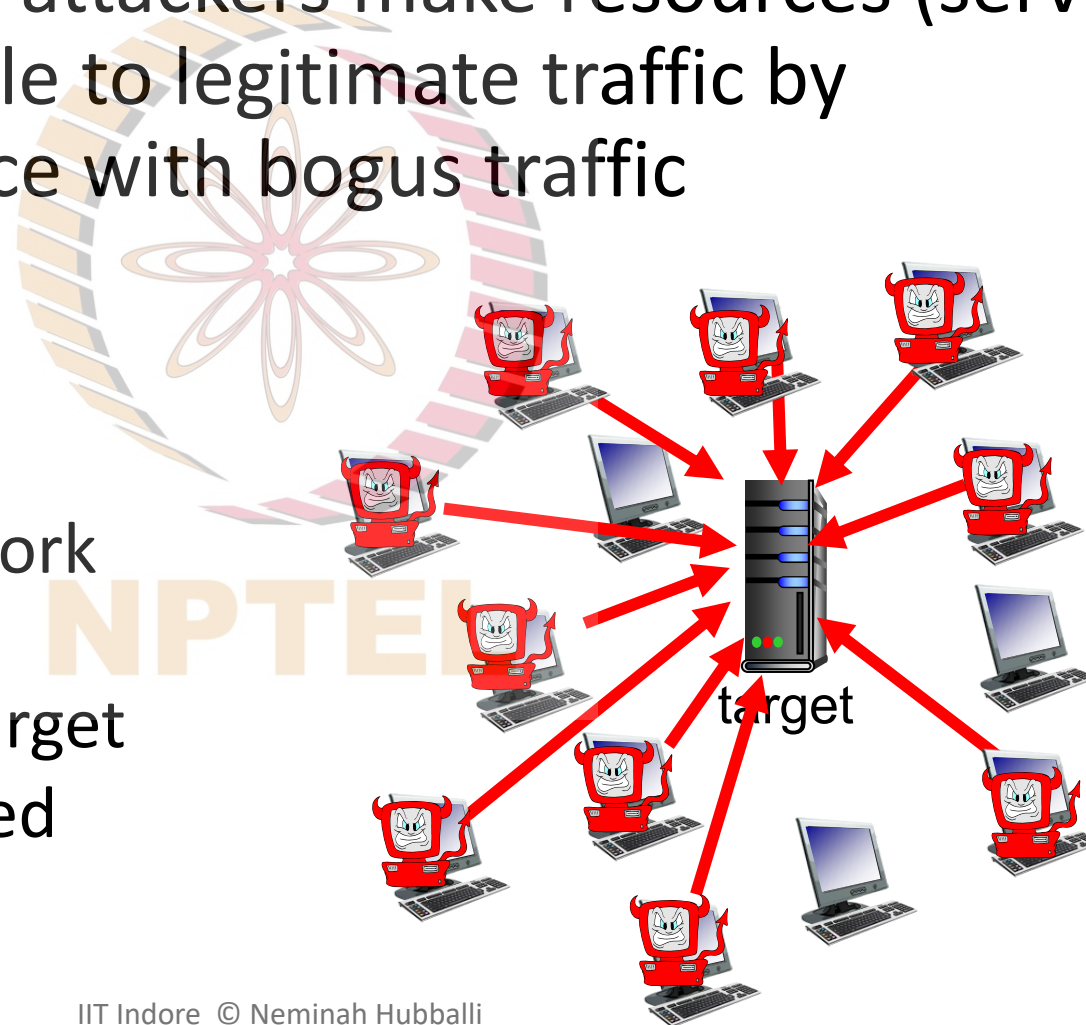
IP spoofing: injection of packet with false source address



Bad guys: Denial of Service

Denial of Service (DoS): attackers make resources (server, bandwidth) unavailable to legitimate traffic by overwhelming resource with bogus traffic

1. select target
2. break into hosts around the network (see botnet)
3. send packets to target from compromised hosts



Lines of Defense:

- ❑ **Authentication:** proving you are who you say you are
 - ❑ cellular networks provides hardware identity via SIM card; no such hardware assist in traditional Internet
- ❑ **Confidentiality:** via encryption
- ❑ **Integrity checks:** digital signatures prevent/detect tampering
- ❑ **Access restrictions:** password-protected VPNs
- ❑ **Firewalls:** specialized “middleboxes” in access and core networks:
 - ❑ off-by-default: filter incoming packets to restrict senders, receivers, applications
 - ❑ detecting/reacting to DOS attacks